Convex optimization for machine learning – TD2

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1 Exercises

Exercise 1. (Proximal operator) Let $f : \mathbb{R}^n \to \mathbb{R} \cup +\infty$ be a convex lower-semicontinuous function such that $\mathrm{dom} f \neq \emptyset$.

1. Recall the definition of the domain of a convex function.

2. It is possible to prove (but we do not ask you to do it) that $\exists \mathbf{x}_0 \in \mathrm{dom} f$ such that $\partial f(\mathbf{x}_0) \neq \emptyset$. Using this information, show that there exists $\alpha \in \mathbb{R}$ and $\mathbf{w} \in \mathbb{R}^n$ such that for all $\mathbf{x}$ we have $f(\mathbf{x}) \geq \alpha + \langle \mathbf{w}, \mathbf{x} \rangle$.

3. Let us fix $\mathbf{x} \in \mathbb{R}^n$. Let us define $g : \mathbf{y} \mapsto f(\mathbf{y}) + \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|^2$. Show that g is strongly convex.

4. Show that $\lim\limits_{\|\mathbf{y}\| \to \infty} g(\mathbf{y}) = +\infty$.

5. Show that g has a minimizer and that it is unique. We will denote this minimizer as $\mathrm{prox}_f(\mathbf{x})$. The function $\mathrm{prox}_f : \mathbb{R}^n \to \mathbb{R}^n$ is called the *proximal operator* of f.

Exercise 2. Let us denote:

$$\mathrm{prox}_g(\mathbf{y}) = \arg\min_{\mathbf{x} \in \mathcal{X}} \Big\{ g(\mathbf{x}) + \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|^2 \Big\}$$

the proximal operator of g at $\mathbf{y}$. Fix $\gamma > 0$. Show that if f and g are convex, and f differentiable, then the fixed points of the nonlinear equation

$$\mathbf{x} = \mathrm{prox}_{\gamma g}\big(\mathbf{x} - \gamma \nabla f(\mathbf{x})\big) \tag{1}$$

are the minimizers of the function $F = f + g$.

Exercise 3. (Taylor-Lagrange inequality) The goal of this exercise is to prove Taylor-Lagrange inequality. This is a fundamental inequality for the study of gradient descent and related methods.

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable function whose gradient is L-Lipschitz *i.e.* $\|\nabla f(\mathbf{y}) - \nabla f(\mathbf{x})\| \leq L\|\mathbf{y} - \mathbf{x}\|$ for all $\mathbf{x}, \mathbf{y}$.

1. Prove that for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, $\big\langle \nabla f(\mathbf{y}) - \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \big\rangle \leq L\|\mathbf{y} - \mathbf{x}\|^2$.

2. Set $\varphi(t) = f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))$ for all $t \in [0, 1]$. Prove that:

$$f(\mathbf{y}) - f(\mathbf{x}) - \big\langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \big\rangle = \varphi(1) - \varphi(0) - \varphi'(0).$$

3. Deduce that:

$$f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle = \int_0^1 \langle \nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \mathrm{d}t$$

4. Using the first question, conclude that

$$f(\mathbf{y}) \leq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle + \frac{L}{2} \|\mathbf{y} - \mathbf{x}\|^2.$$

2 Solutions

Exercise 1 1. The domain of the function f is the set of points $\mathbf{x} \in \mathbb{R}^n$ such that $f(\mathbf{x}) \neq +\infty$.

2. By definition of the sub-differential, we have for $\mathbf{q}_0 \in \partial f(\mathbf{x}_0)$:

$$\forall \mathbf{x} \in \mathbb{R}^n, f(\mathbf{x}) \geq f(\mathbf{x}_0) + \langle \mathbf{q}_0, \mathbf{x} - \mathbf{x}_0 \rangle = \big[f(\mathbf{x}_0) - \langle \mathbf{q}_0, \mathbf{x}_0 \rangle\big] + \langle \mathbf{q}_0, \mathbf{x} \rangle.$$

Setting $\alpha := f(\mathbf{x}_0) - \langle \mathbf{q}_0, \mathbf{x}_0 \rangle$ and $\mathbf{w} = \mathbf{q}_0$, we obtain the desired result.

3. By definition, a function h is strongly convex if $\mathbf{y} \mapsto h(\mathbf{y}) - \frac{\mu}{2}\|\mathbf{y}\|^2$ is convex for some $\mu \geq 0$. Since

$$g(\mathbf{y}) = \Big[f(\mathbf{y}) - \langle \mathbf{x}, \mathbf{y} \rangle + \frac{1}{2}\|\mathbf{x}\|^2\Big] + \frac{1}{2}\|\mathbf{y}\|^2,$$

and the function f is convex, as well as the affine function $\mathbf{y} \mapsto -\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{x}\|^2/2$, it is clear that g is 1-strongly convex.

4. By definition of g, and by using question 2, we have:

$$g(\mathbf{y}) \geq \alpha + \langle \mathbf{w}, \mathbf{y} \rangle + \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|^2.$$

This inequality proves that g converges to $+\infty$ when $\|y\|$ goes to infinity (since $\langle \mathbf{w}, \mathbf{y} \rangle \geq -\|\mathbf{w}\|\|\mathbf{y}\|$ by Cauchy-Schwarz inequality).

5.
- **(Existence)** The existence of a minimizer $\mathbf{x}^*$ of the l.s.c. and coercive function g is a consequence of the results seen during the lecture.
- **(Unicity)** We will prove a general and important result which is useful to keep in mind: a strongly convex function h admits at most one minimizer. Assume the converse. There exist distinct $\mathbf{x}^*, \mathbf{x}^{**}$ such that:

$$h(\mathbf{x}^*) = h(\mathbf{x}^{**}) = \min_{\mathbf{x} \in \mathcal{X}} h(\mathbf{x}).$$

Let $t \in]0, 1[$ (any t in this interval will do, you can assume $t = 1/2$ will do if you wish[1]). Since h is μ-strongly convex for some $\mu > 0$, we have by the very definition of the convexity of $\mathbf{x} \mapsto h(\mathbf{x}) - \mu\|\mathbf{x}\|^2/2$:

$$\begin{aligned} h\big(t\mathbf{x}^* + (1-t)\mathbf{x}^{**}\big) &\leq t h(\mathbf{x}^*) + (1-t)h(\mathbf{x}^{**}) - \frac{\mu}{2}\Big[\|t\mathbf{x}^* + (1-t)\mathbf{x}^{**}\|^2 - t\|\mathbf{x}^*\|^2 - (1-t)\|\mathbf{x}^{**}\|^2\Big] \\ &= h(\mathbf{x}^*) - \frac{\mu}{2}\Big[\|t\mathbf{x}^* + (1-t)\mathbf{x}^{**}\|^2 - t\|\mathbf{x}^*\|^2 - (1-t)\|\mathbf{x}^{**}\|^2\Big] \\ &< h(\mathbf{x}^*) \\ &\leq h\big(t\mathbf{x}^* + (1-t)\mathbf{x}^{**}\big), \end{aligned}$$

since the term within the brackets is positive by (strict) convexity of the squared Euclidean norm function. The last inequality stems from the fact $\mathbf{x}^{**}$ is a minimizer of h. But it is absurd! So h can only have one minimizer at most. This result proves that g admits exactly one minimizer.

This minimizer depends on $\mathbf{x}$ and since it is unique, it defines a function of $\mathbf{x}$: the proximal operator of f at point $\mathbf{x}$.

[1] But it's more annoying to write!

Exercise 2 By assumption, $\mathbf{x} = \mathrm{prox}_{\gamma g}\big(\mathbf{x} - \gamma\nabla f(\mathbf{x})\big)$, so it minimizes the function:

$$\varphi : \mathbf{y} \mapsto \gamma g(\mathbf{y}) + \frac{1}{2}\big\|\mathbf{y} - \big(\mathbf{x} - \gamma\nabla f(\mathbf{x})\big)\big\|^2$$

By Fermat's rule, this implies that $\mathbf{0} \in \partial\varphi(\mathbf{x})$. Let's compute this sub-gradient.

$$\begin{aligned}
\partial\varphi(\mathbf{x}) &= \partial(\gamma g)(\mathbf{x}) + \big\{\mathbf{x} - \big(\mathbf{x} - \gamma\nabla f(\mathbf{x})\big)\big\} \\
&= \partial(\gamma g)(\mathbf{x}) + \big\{\gamma\nabla f(\mathbf{x})\big\} \\
&= \gamma\big(\partial g(\mathbf{x}) + \big\{\nabla f(\mathbf{x})\big\}\big) \\
&= \gamma\partial(f+g)(\mathbf{x}),
\end{aligned}$$

since $\gamma \geq 0$ (this justifies that $\partial(\gamma g)(\mathbf{x}) = \gamma\partial g(\mathbf{x})$). And so

$$\mathbf{0} \in \partial(f+g)(\mathbf{x}).$$

Therefore, the reciprocal of Fermat's rule ensures that $\mathbf{x}$ is a minimizer of $f+g$.

Exercise 3

1. Cauchy-Schwarz inequality and the assumption on f give the desired inequality:

$$\big\langle\nabla f(\mathbf{y}) - \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x}\big\rangle \leq \|\nabla f(\mathbf{y}) - \nabla f(\mathbf{x})\|\|\mathbf{y} - \mathbf{x}\| \leq L\|\mathbf{y} - \mathbf{x}\|^2.$$

2. It is clear that, by definition, $\varphi(0) = f(\mathbf{x})$ and $\varphi(1) = f(\mathbf{y})$. We must now compute the derivative of φ at $t = 0$. Notice that $\varphi = f \circ A$, where A is the affine map from $[0,1]$ to $\mathbb{R}^n$ defined by $A(t) = \mathbf{x} + t(\mathbf{y} - \mathbf{x})$. The chain rule gives:

$$\begin{aligned}
\forall h \in \mathbb{R},\ \varphi'(t)h &= \mathrm{d}\varphi(t).h \\
&= \mathrm{d}f\big(A(t)\big).\big(\mathrm{d}A(t).h\big) \\
&= \mathrm{d}f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})).\big(h(\mathbf{y} - \mathbf{x})\big) \\
&= h\mathrm{d}f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})).(\mathbf{y} - \mathbf{x}) \\
&= h\big\langle\nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})), \mathbf{y} - \mathbf{x}\big\rangle,
\end{aligned}$$

   by linearity of the differential. Simplifying by h and taking $t = 0$, we get $\varphi'(0) = \big\langle\nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x}\big\rangle$ as expected.

3. By the fundamental theorem of calculus applied to φ, we have:

$$\varphi(1) - \varphi(0) = \int_0^1 \varphi'(t)\mathrm{d}t = \int_0^1 \big\langle\nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})), \mathbf{y} - \mathbf{x}\big\rangle\mathrm{d}t.$$

   Since $\varphi'(0) = \int_0^1 \big\langle\nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x}\big\rangle$ thanks to the previous question, we get the result.

4. It suffices to use the inequality of the first question in the integral of the previous question to obtain that

$$f(\mathbf{y}) - f(\mathbf{x}) - \big\langle\nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x}\big\rangle \leq L\|\mathbf{y} - \mathbf{x}\|^2 \int_0^1 \frac{t^2}{t}\mathrm{d}t = \frac{L}{2}\|\mathbf{y} - \mathbf{x}\|^2.$$